

6423 Project Proposal

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Incremental Indexing for TokenSmith

Problem.

TokenSmith currently rebuilds its entire retrieval index from scratch whenever documents are indexed. This is slow and wasteful, especially for large textbook-based corpora.

Why it matters.

A local RAG system is only practical if indexing is efficient. Full re-indexing increases latency, wastes compute, and makes it harder to update the corpus incrementally.

Approach.

I will add incremental indexing at the document level. The system will maintain a manifest with file hashes and artifact metadata, detect which source documents are new or changed, and only re-run parsing, chunking, and embedding for those files. Unchanged artifacts will be reused when rebuilding the global index.

Validation.

I will verify that incremental indexing produces the same final index contents and query behavior as a full rebuild. I will test add, modify, and delete cases.

Performance evaluation.

I will compare full vs. incremental indexing on:

- indexing time
- number of documents/chunks reprocessed
- amount of embedding work avoided

Resources needed.

TokenSmith codebase, provided textbook PDF(s), local GGUF models, Python/Conda environment, FAISS, and BM25.

Goals.

- 75%: Detect unchanged documents and skip reprocessing them.
- 100%: Store per-document artifacts and rebuild the merged index correctly from changed files only.
- 125%: Extend the design to section-level or chunk-level incremental indexing and evaluate the tradeoffs.